Ayush Mishra

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Summary

Software Test Engineer with an MS in Computer Science and 3+ years of experience bridging enterprise network validation, scalable backend development, and AI integration. Proven expertise at validating complex control-plane features and engineering scalable Python/Pytest automation frameworks using hardware traffic generators (Ixia). Passionate about combining full-stack development with rigorous infrastructure testing to ensure zero-defect network deployments.

Education

University of Colorado-Boulder **May 2026**
MS, Computer Science (Network Engineering and Backend Development) *Boulder, CO*

Relevant Course Work: Enterprise Networks, Network Systems, Network Automation, IP Routing and Protocols, Network Analysis, and Modeling

University of Nebraska-Lincoln **May 2021**
BS, Computer Science (Minors: Physics, Philosophy) *Lincoln, NE*

Capabilities

- Proficient in designing and implementing IP addressing schemes, Ethernet technologies, and enterprise routing protocols.
- Adept at utilizing packet analysis tools (Wireshark, TCPdump) for in-depth protocol analysis, fault isolation, and troubleshooting.
- Experienced in executing feature, scale, and stress testing using hardware traffic generators (Ixia) to validate network resilience under line-rate conditions.
- Capable of developing and maintaining scalable automated test suites and network provisioning workflows using Python, Pytest, Netmiko, and NAPALM.
- Hands-on experience deploying security protocols, including IPSec-based VPN services, for secure enterprise connectivity.
- Skilled in full-stack UI application development, scalable API design, and database architecture for high-transaction systems.
- Strong command of version control and software development workflows using Git and BitBucket.
- Experienced in integrating Agentic LLMs and AI technologies into modern software and network automation frameworks.

Network Experience

Arista Networks **July 2026 – Pres.**
Software(Network) Test Engineer *Santa Clara, CA*

- Owned testing and validation for critical Arista EOS features, performing rigorous protocol analysis, test planning, and technical knowledge transfer (TOIs)
- Engineered scalable automated test suites using Python and Pytest, leveraging hardware traffic generators (Ixia) to validate system resilience under line-rate network stress.
- Debugged complex vulnerabilities and packet-drop anomalies during edge-case scale testing, collaborating directly with core developers to resolve critical defects prior to release.

Network Deployment

- At least 300 hours of hands-on experience implementing the following:
  - MPLS with OSPF/LDP and MPLS-TE (RSVP-TE/CSPF) for primary/backup path engineering, and Policy-Based Routing.
  - Campus-style Layer 2/3 switching with VLANs, trunking, inter-VLAN routing, and enhanced resiliency using STP/RSTP and EtherChannel; Also IP resiliency using HSRP and ECMP.

- Deployment of various routing protocols (OSPF, RIP, EIGRP, BGP) for Enterprise and Internet uplink connectivity; and relevant features needed for private to public network communication.
- Developed software applications for network services in C, including a UDP-based FTP client/server, a concurrent TCP HTTP web server, an HTTP caching and reverse proxy, and a distributed file system that shards files across multiple servers for redundancy.

Network Automation

- Well versed in the use of network automation tools including:
  - Automated multi-router configuration workflows using Python (Netmiko/NAPALM) with external config files, IPv4 validation + reachability checks, and saved device running-config snapshots for repeatable deployments.
  - Deployed network automation tools to collect routing protocol inputs, configuration snapshot storage in database, and generate config diffs to detect drift between current and previously saved configurations.
  - Integrate LLMs/AI in automation applications for various network configuration tasks.

Software Development Experience

Cisco Systems *(Capstone)* **Sep 2025 – May 2026.**
Software (Network) Engineer *Boulder, CO*

- Development of an Agentic LLM-driven, intent-based network configuration system, using Python/Flask backend, MongoDB vector search, and React UI.
- Infused Prompt-Engineering, LangChain, Token limitation, and custom tool calling into the application to perform network automation tasks using user prompts.

VTM Group **May 2023 – Aug 2024**
Technical Marketing Engineer *Beaverton, OR*

- Worked on centralized platforms to collect and manage test results from various hardware vendors.
- Contributed with development of tools used for testing for PCI-SIG.

Novus Labs **Oct 2021 – Apr 2023**
Software Developer *Hillsboro, OR*

- Worked to build solutions for Fortune 500 clients using C++ and Python.
- Built backend for Web/IoT Applications for real-time data storage, and tools for Law-firm using Pacer-API.
- Also, implemented source-to-source code transformation using clang-LibTooling (C++) for ROS.

Tokn Inc. *(Startup)* **May 2021 – Jan 2022**
Software Developer (Backend) *Remote*

- Designed backend to use blockchain technologies for the commercialization of music, using NoSQL and Solidity.

Technologies & Tools

- **Programming Languages:** C++; Python; Cisco IOS; JS (ReactJs, VueJs); MongoDB; SQL; AWS; Docker; Linux; Laravel
- **Network Protocols:** IPv4/IPv6; TCP/IP; OSPF; BGP; ARP; DNS; STP; RIP; EIGRP; Wi-Fi; MPLS; IPSec; QoS
- **Network Automation:** Netmiko; NAPALM; Wireshark; GNS3; Jinja2; Ansible; Container Lab; IaC; IXIA
- **AI Software:** LangChain; RAG; Prompt Engineering; Fine-Tuning; MCP
- **Version Tools:** Git; BitBucket; Jira

Certifications

- **Cisco Certified Network Associate** (CCNA, Exp. Dec 2026)
- **JNCIS DevOps** (Exp. Dec 2026)